

Streamlining 4D Cardiac Image Workflows: Open-Source Tools for Segmentation, Registration, and Visualization

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Abstract. 4D cardiac imaging offers impactful insights into structural heart dynamics. However, analyzing 4D data using conventional 3D software tools requires additional effort. This paper introduces three natively 4D-optimized software applications we have developed: *ITK-SNAP 4* for 4D image I/O, visualization, and segmentation; *Greedy Propagation* for intra-series registration and creation of 4D segmentation from sparse 3D segmentations; and *Scherzo* for fast web-based 4D model generation and visualization. These open-source tools provide a streamlined user experience, comprehensive features, and broad file type support including common 4D cardiac image formats. Experiments on core features demonstrate feasibility and consistency.

Keywords: 4D Cardiac Image Analysis · 4D Segmentation · Intra-Series Registration · Web-Based Visualization · Open-Source Software · ITK-SNAP 4 · Greedy Propagation · Scherzo.

1 Introduction

1.1 The Need for 4D Support in Cardiac Image Analysis

4D cardiac imaging, which adds a time dimension to 3D volumes (3D + T), has become increasingly common in diagnosing cardiac diseases[7], planning surgeries[10], and studying heart function[24,22]. By capturing dynamic changes over time, 4D imaging provides critical insights into heart structure and function that are difficult to obtain with 2D images or static 3D volumes[8]. However, 3D software tools often fall short in handling 4D data efficiently. Users are frequently forced to extract and process individual 3D volumes from a 4D file, only to reassemble them later—a workflow that is cumbersome, inefficient, and error prone. Moreover, many tools lack essential features tailored for cardiac analysis, such as support for cardiac-specific file formats (e.g., 4D transesophageal echocardiography (TEE) and 4D computed tomography (CT) image series) or intuitive navigation of the time dimension. This underscores the need for software solutions optimized for seamless 4D cardiac image analysis.

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1.2 4D Image I/O, Viewing, and Segmentation

The core components of any cardiac image analysis workflow—image I/O, viewing, and segmentation—are well-supported for 3D data by existing open-source tools. However, with the time dimension added, working with 4D images is not an intuitive and integrated experience using 3D analysis software. There are a few examples of open-source software that address this issue. *SlicerHeart* [19] introduced extensions for *3D Slicer* [23] that enable I/O, viewing and segmentation of 4D data. However, the 3D nature of the Slicer platform makes 4D analysis dependent on extensions. Similarly, *Horos* [1] and *Osirix* [25] offer 4D viewing but have limited support for 4D file formats, are platform-restricted to macOS, and do not include segmentation features.

ITK-SNAP [30] is an open-source cross-platform software application for biomedical image segmentation. It was created to be easy-to-use for users from diverse backgrounds. Since its release in the early 2000s, it has had more than 940,000 downloads globally [2] and has evolved into a versatile tool that is compatible with a wide range of image modalities and file formats. Starting with version 4.0, the application introduced comprehensive support for 4D data, enhancing its segmentation capabilities and adding features tailored for cardiac analysis. This paper highlights these advancements and demonstrates their integration into efficient 4D cardiac analysis workflows.

1.3 Intra-Series Registration Tools

4D segmentation is a crucial component in cardiac image analysis, especially for modeling dynamics of the chambers, valves, and vasculature. While fully automated cardiac image segmentation has advanced rapidly, manual and semi-automated segmentation remain essential for validation studies and for tasks that lack fully automated solutions. However, manual tracing of heart structure remains a time-consuming task, and it is common for only a few time points to be manually traced in a study. An approach to address this challenge involves applying image registration to "propagate" a manual segmentation across time points, creating a 4D segmentation from 3D data.

Various registration frameworks can be used for intra-series registration, among which, methods such as *Elastix* [18], *ANTs* [27], and *FireANTs* [14] can provide accurate results with higher computation cost and potential GPU requirements, while deep learning based methods such as *VoxelMorph* [6] and *DeepReg* [12] can run extremely fast once trained, but require extensive training and GPU resources. *Greedy* is a C++ implementation of the greedy diffeomorphic registration algorithm [31]. As a cross-platform CPU-based application, *Greedy* offers portability, ease of use, and integration with *ITK-SNAP* both through its user interface and as a companion tool. An intra-series registration pipeline based on Greedy was introduced to propagate a 3D volumetric segmentation to other time points within the same image series, as demonstrated in several applications [24, 4]. This paper introduces *Greedy Propagation*, a C++ implementation of that pipeline integrated as one of the *Greedy* executables with improved efficiency and configurability.

1.4 Web-Based 4D Modeling and Visualization Tools

Effective communication of segmentation results requires creation and visualization of high-quality 4D models that can be leveraged for downstream tasks, such as measurement computation and finite element analysis. However, setting up local software environments can hinder result sharing and collaboration. Web-based solutions offer an ideal alternative, providing portability and accessibility across devices and operating systems. Existing web-based tools, such as *Volview* [28], *Glance* [15], and *ITK-VTK-Viewer* [16], address some visualization needs but lack support for 4D data or essential features like surface mesh visualization. To fill these gaps, this paper introduces *Scherzo*, a minimalist, web-based application for 4D model building and visualization. *Scherzo* prioritizes simplicity and user experience, targeting broad accessibility for cardiac researchers.

Comprehensive 4D cardiac analysis involves multiple stages, from image acquisition and pre-processing to segmentation, modeling, and result delivery. Researchers often experiment with various tools to optimize their workflows, making portability, compatibility, and configurability crucial [29,9]. In the experiment design section, this paper presents case studies demonstrating how the proposed toolset—*ITK-SNAP 4*, *Greedy Propagation*, and *Scherzo*—can be integrated to create flexible and efficient 4D cardiac analysis pipelines.

2 Methods

The research team led the development, testing, and integration of the proposed open-source software tools. These tools were specifically designed and optimized to support efficient 4D cardiac image analysis. Details of each tool’s functionality and role within the study are provided in the following sections.

2.1 ITK-SNAP 4: Enhancements for 4D Cardiac Image Analysis

In *ITK-SNAP 4*, the core data structure and application framework were upgraded to fully support 4D data. The application now utilizes the 4D `itk::Image` data structure [21] to represent images, overlays, and segmentations. By referencing individual time-point volumes, this design ensures rapid access without additional memory overhead. Users can now seamlessly perform segmentations with all existing tools directly in 4D, eliminating the need for intermediate file operations. Furthermore, *ITK-SNAP 4* incorporates the `TDigest` data structure [11] to efficiently compute histograms and approximate image quantiles in large 4D datasets, enhancing contrast adjustment capabilities.

The updated framework also introduces UI enhancements (Fig. 1), including dynamic adaptations for 4D data and shortcuts for efficient time-point navigation. Unlike *ITK-SNAP 3*, which stores 4D images as multi-component images and does not support multi-component segmentations, the new framework allows segmentations at each time point. When a 4D image is loaded, a fourth crosshair dimension appears in the navigation panel (Fig. 1b), and a time point indicator

(current/total) is displayed on each slicing window (Fig. 1c). The Layer Inspector now includes multiple 4D-specific tools: The Time Point Properties panel (Fig. 1d) supports 4D replay and time-point-wise annotation; The Image Info Panel (Fig. 1e) displays additional information regarding the time dimension, such as the intensity-under-cursor across all time points.

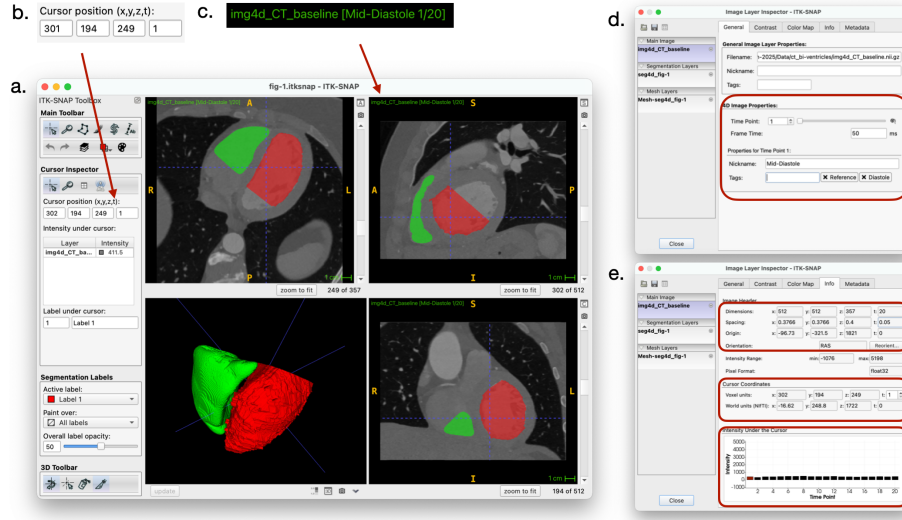


Fig. 1. *ITK-SNAP 4* User Interface: a) Main Image Window, with a 4D CT base layer and a Bi-Ventricular segmentation layer loaded; b) 4D Navigation Panel; c) In-window Time Point Information; d) Layer Inspector with 4D Properties highlighted; e) Layer Inspector Info Panel with 4D elements highlighted.

ITK-SNAP 4 also extends support from 2D and 3D image file types to common 4D formats and, importantly, includes support for several common file types specifically for cardiac imaging, such as 4D Echo Cartesian Dicom (Philips), 4D CTA Dicom Series (GE and Siemens), and NRRD Volume Sequence (seq.nrrd) from *3D Slicer*. Externally generated anatomical meshes can now be loaded, and multi-label surface meshes can be exported as STL files for 3D printing. Additionally, the Image Free Rotation Tool (Fig. 2), introduced in version 4.2, allows interactive rotation of image planes without reslicing. All segmentation tools are available on rotated images, with rasterization applied when using tools such as the polygon and paintbrush tools.

2.2 Greedy Propagation: Generating 4D Segmentations from 3D Data

Greedy Propagation is a command-line application built on the existing *Greedy Registration API*, designed to enable the creation of 4D segmentations from

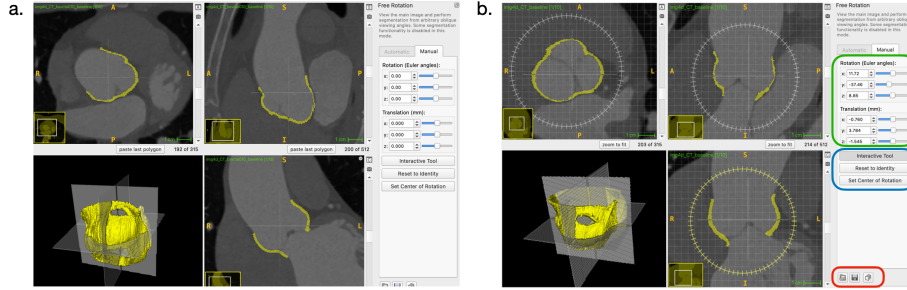


Fig. 2. *ITK-SNAP 4* Free Rotation Tool example with a 4D CT base layer and an Aortic Root segmentation layer loaded: a) Workspace without rotation; b) Workspace with rotation. The UI includes an Interactive Rotation Widget (white dial), rotation and translation sliders (green), buttons for activating the Interactive Widget (blue), and buttons for loading/saving a transformation file, and image reslicing (red).

sparse 3D segmentation(s) using multiple steps of deformable registration[4]. Greedy Propagation has two user interfaces, a Command-Line Interface (CLI) and an Application Programming Interface (API). It requires the following inputs: 1) a 4D image; 2) a reference 3D segmentation for a specific time point; 3) the time point of the reference segmentation; 4) a list of target time points to which the reference segmentation will be warped. The outputs include: 1) a 4D segmentation file containing warped segmentations for all target time points; 2) a series of 3D segmentation images corresponding to the target time points; 3) segmentation meshes for the target time points. Users can also include additional meshes of the reference time point to be warped to the target time points, with mesh topology preserved between different time points. In addition to the required inputs, optional parameters can be configured for fine-tuning the propagation or experiments.

2.3 Scherzo: A Web-based 4D Model Generator and Viewer

Leveraging the latest advancement in web-based medical image processing and visualization[20,26,17], *Scherzo*[3], a web application, is developed for quick and interactive 4D surface mesh model creation and visualization. Users can access the application by entering the URL in a web browser. Once loaded, the application runs entirely on the user’s device, requiring no registration or login. Upon launch, users are presented with a simple interface to upload a 4D segmentation image (Fig. 4a). After clicking the “Generate Model” button, the application processes the input and generates a 4D model (Fig. 4b) using marching cubes. Once the model is created, a 4D model viewer is displayed, featuring a control panel with rendering options, navigation tools, and export capabilities (Fig. 4c).

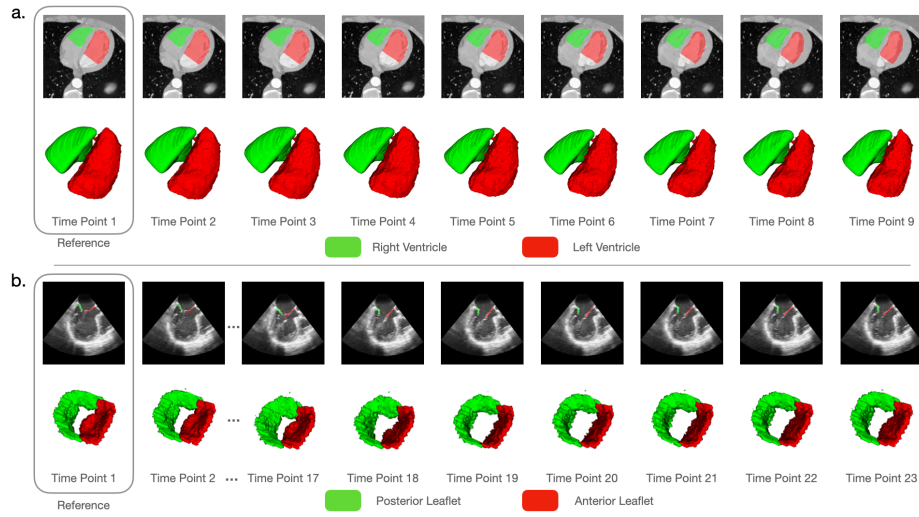


Fig. 3. *Greedy Propagation* Results. For each study, nine time-points are selected for display, with segmentations overlaid on gray-scale images. Segmentations were propagated from the reference time point to all the other time points: a) CT Bi-Ventricle; b) TEE Mitral Valve.

2.4 Experiment Design

The experiments are designed to evaluate the effectiveness of the primary features of each tool and to demonstrate the feasibility of integrating these tools into a cohesive 4D analysis workflow. Five studies were selected to represent a variety of imaging modalities and segmented cardiac structures: 1) a TEE study of the mitral valve; 2) a TEE study of the left ventricle; 3) a TEE study of the aortic valve; 4) a CT study of the left and right ventricles; 5) a CT study of the aortic root. All studies are in cardiac-specific 4D file formats (4D Echo Cartesian Dicom and 4D CTA Dicom Series) that were previously not supported by *ITK-SNAP* 3, and were read using added I/O logic in *ITK-SNAP* 4.

For each study, the target cardiac structures were manually segmented at two separate time points using *ITK-SNAP* 4: one time point was designated as the reference for *Greedy Propagation*, and the other as the target. For studies 1, 4, and 5, where the anatomical structures were not aligned with the three orthogonal image axes during acquisition, each reference segmentation was performed twice—once with free rotation enabled in *ITK-SNAP* and once without. *Greedy Propagation* was then applied to generate a 4D segmentation from the 3D segmentation of the reference time point. The resulting 4D segmentations were subsequently imported into *ITK-SNAP* and Scherzo for surface mesh generation. For study 5, an additional manual segmentation was produced by a different tracer and was propagated for inter-observer variability analysis. Multiple

target time points were manually traced by the same observer for comparison with the propagated segmentations.

3 Results

3.1 Free rotation tool in ITK-SNAP 4

For the two segmentations performed on the reference time point, the time spent with and without rotation were recorded and compared. An average reduction of 43.5% in tracing time shows the effectiveness of the free-rotation tool when tracing images where the structures are not aligned with the slicing axes during acquisition.

Table 1. Tracing time with/without free-rotation tool using *ITK-SNAP*.

Modality	Structure	Tracing Time		Manual Tracing
		without rotation	with rotation	Efficiency Improvement
CT	Aortic Root	1 hr 20 min	30 min	62.5%
CT	Bi-Ventricles	4 hr 10 min	2 hr 40 min	36%
TEE	Mitral Valve	25 min	17 min	32 %
Average				43.5%

3.2 Greedy propagation

For each study, the propagated segmentation of the target time point was compared to the manually traced target segmentation, and the Mean Symmetric Distance (MSD) was recorded (Table 2). The result shows the MSDs between the propagated and the manually traced segmentations are comparable to the inter-observer variability reported in previous cardiac segmentation studies[13,5]. For study 5, the MSDs between propagation results from two different tracer’s manual segmentations were recorded in Fig. 5.

3.3 Scherzo Model Generation

The models generated by *Scherzo* were validated by comparison with models generated by *ITK-SNAP*, as they use slightly different methods of smoothing and decimation logic due to differences in library availability between web and desktop environments. Two sets of models, one by *Scherzo* and one by *ITK-SNAP*, were generated from 4D segmentation in five 4D image studies and were compared using the mesh distance metric. The result (Table 2) shows models generated from *Scherzo* were comparable to models generated from *ITK-SNAP*.

A browser and screen compatibility test for *Scherzo* was conducted across four browsers (Chrome, Safari, Edge, and Firefox) and three device types: Desktop (MacBook Pro 2016, Intel CPU, 16 GB RAM), Tablet (iPad Pro 6th Generation), and Mobile (iPhone 15 Plus). *Scherzo* successfully ran on all tested

Table 2. Mean Symmetric Distance (MSD) between propagated time-point volume and manually traced time-point volume, and between Scherzo generated models and ITK-SNAP generated models.

Modality Structure		Num Frames Between Reference & Target Time Points	MSD Between Propagated & Traced Segmentation (mm)	MSD Between <i>Scherzo</i> & <i>ITK-SNAP</i> Models (mm)
CT	Aortic Root	7	0.41 ± 0.06	0.05 ± 0.02
CT	Bi-Ventricles	3	2.94 ± 0.93	0.03 ± 0.01
TEE	Aortic Valve	31	1.17 ± 0.35	0.03 ± 0.01
TEE	Left Ventricle	7	2.11 ± 0.16	0.03 ± 0.01
TEE	Mitral Valve	22	0.74 ± 0.28	0.03 ± 0.01

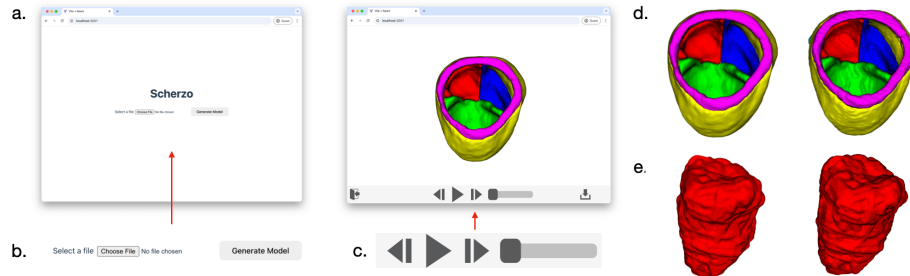


Fig. 4. Scherzo UI and results: a) Left: Welcome Page; Right: 4D Model Viewer; b) File Selection Panel in the Welcome Page; c) 4D Replay Panel in the Viewer; d) Aortic Valve Result: Left: *Scherzo*; Right: *ITK-SNAP*; e) Left-Ventricle Result: Left: *Scherzo*; Right: *ITK-SNAP*.

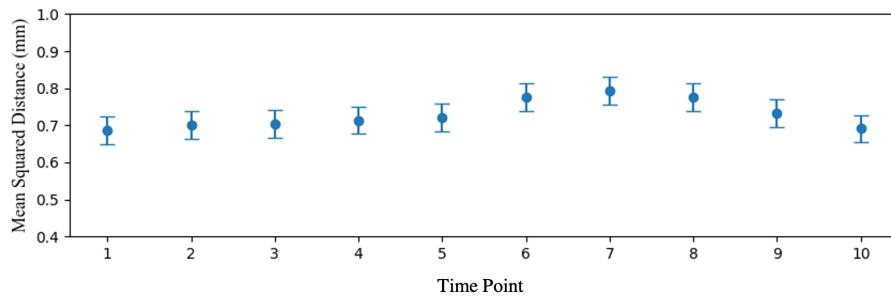


Fig. 5. Inter-Observer Variability in Greedy Propagation: Mean Symmetric Distance (MSD) between propagation result from two different tracers' input were plotted for each time point.

browser-device combinations without compatibility issues. Page loading time and model generation performance were evaluated in the desktop environment, where the front page consistently loaded in under 1.5 seconds on average. A detailed breakdown of model generation performance is presented in Table 3.

Table 3. Scherzo model generation performance

Modality Structure		Frame Size	Num Labels	Model Computation Time in Seconds
CT	Aortic Root	$512 \times 512 \times 315$	1	16.28
CT	Bi-Ventricles	$384 \times 384 \times 268$	2	36.19
TEE	Aortic Valve	$224 \times 144 \times 208$	6	7.65
TEE	Left Ventricle	$224 \times 208 \times 208$	1	3.97
TEE	Mitral Valve	$224 \times 208 \times 208$	2	4.27

4 Discussion

This paper presents software applications that make several key contributions to 4D cardiac image analysis. First, the comprehensive 4D upgrade in *ITK-SNAP 4* enables the reading of 4D cardiac specific image types, such as 4D TEE from iE33 and EPIQ platforms and 4D CT from Siemens and GE scanners, allowing seamless integration of these data into existing analysis workflows. Second, the introduction of the Free-Rotation Tool significantly improves the efficiency in manual segmentation, reducing the effort required for tracing anatomical structures that are misaligned with the image axes during acquisition. Third, *Greedy Propagation* command-line tool and API provide a semi-automated approach for generating 4D segmentation from sparse 3D manual segmentations, reducing time spent in manual tracing from tens of hours to under an hour for a typical 4D cardiac sequence with approximately 20 image volumes. Lastly, *Scherzo* provides a fast, lightweight, and highly portable web-based solution for generating and visualizing 4D label image models, making it easy to present results and share them in collaborative environments. In sum, the experimental results demonstrate that these tools deliver effective and consistent outputs while integrating seamlessly with one another and other commonly used software, enabling flexible and complex workflows. The proposed tools prioritize efficiency, user-friendliness, and broad image format compatibility. These characteristics make the tools particularly valuable for researchers and clinicians who require a smooth learning curve and effortless integration into existing pipelines.

Compared to widely used platforms such as *3D Slicer* and its *SlicerHeart* extension, the proposed tools prioritize a streamlined user experience tailored for specialized tasks in 4D cardiac imaging, which enhances accessibility but limits flexibility for more extensive features and customization. *ITK-SNAP 4*

and *Scherzo* emphasize intuitive interfaces designed for efficient 4D segmentation, model building, and visualization, whereas *3D Slicer*, particularly with *SlicerHeart*, provides a feature-rich platform for in-depth analysis, but with a steeper learning curve. On customization: while the proposed tools offer configurable workflows with high-level options, *3D Slicer* enables significantly broader customization through its extensive extension system. In terms of computational performance, the proposed tools focus on portability and efficiency across a wider range of hardware, making them accessible in resource-constrained environments, whereas *3D Slicer*, with its powerful extensions, provides greater opportunities to optimize performance for high-end hardware, including GPUs, which can be advantageous for more computationally intensive workflows. While this qualitative comparison highlights key differences, a more rigorous quantitative benchmarking study across usability, computational performance, and segmentation accuracy is necessary for a comprehensive evaluation, which we identify as an important future direction.

While the proposed tools were initially designed to streamline image analysis workflows for researchers, their integration into clinical practice remains a long-term objective. However, achieving this will require overcoming significant challenges. First, data complexity limits the tools’ ability to provide seamless interoperability with hospital PACS infrastructure, as different imaging vendors encode time-series data in varying ways. Addressing this will require continuous development to ensure broader compatibility, though this remains a complex and resource-intensive endeavor. Additionally, the tools were designed with feasibility in mind, ensuring they can operate efficiently even in environments with limited computational resources. However, the variability in computational capabilities across clinical settings necessitates software solutions that are both highly portable and capable of dynamically optimizing performance based on available hardware. This includes efficiently leveraging GPUs when available to enhance processing speed while maintaining adaptability for resource-constrained environments. Lastly, user training and adoption present additional hurdles, as clinical users may have limited experience with advanced segmentation tools. Overcoming this challenge will require sustained efforts to refine user interfaces, develop intuitive training resources, and tailor workflows that align with real-world clinical demands.

Beyond usability, regulatory compliance represents a long-term challenge in translating research tools into clinical applications. Meeting FDA (U.S.) and CE marking (Europe) regulations require extensive validation, ongoing updates to meet evolving medical standards, and dedicated resources to establish these tools as clinically viable solutions. Furthermore, data privacy and security considerations, particularly in compliance with HIPAA (U.S.) and GDPR (Europe), will demand continuous advancements in secure data management and encryption to align with hospital cybersecurity policies.

Moving forward, several directions will be explored to enhance the usability, compatibility and performance of these tools. First, improving user experience will be a major focus for better accessibility and efficiency. This includes sim-

plifying user interfaces, developing interactive tutorials and improving automation. Additionally, usability studies will be conducted to gather feedback from clinicians and researchers, ensuring modifications align with real-world needs. Second, expanding format compatibility will be prioritized to enhance interoperability with diverse imaging systems and other commonly used software tools. Finally, leveraging GPU acceleration to further improve computational performance will be a goal. This will involve adopting *Fire-ANTs* to accelerate registration performance, and exploring *WebGPU* to enable faster real-time rendering in web-based applications.

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